SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3_AT3 — VISUALIZATION CRITIQUE & REDESIGN

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO3 – Evaluation (BL5)	**Assessment:**	Assessment Tool 3 – Critique & Redesign
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO3 Statement:	*Evaluate flawed or misleading data visualizations, critique their specific shortcomings, and justify an improved redesign.(BL5)*		
Student Name:	M.SAI HARSHITHA	**Reg. No.:**	192424408
Section / Batch:	B. TECH AI & DS	**Date:**	18.08.2026

Code of Conduct

I M.SAI HARSHITHA/192424408 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond simply naming a flaw: explain WHY it is misleading and what a reader could wrongly conclude from it.
- Your redesign must be a specific, justified chart proposal — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION CRITIQUE & REDESIGN

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Visualization Critique & Redesign

CO3 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO3-AT3 Visualization Critique & Redesign problems, in which students are given a flawed or misleading chart, must critique its specific shortcomings, and propose a justified redesign.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the specific visualization principle being violated (e.g., proportional ink, axis scaling, color perception, overplotting) in the given flawed chart.	Good understanding of the relevant principle, with only minor gaps in identifying exactly what is violated.	Basic understanding, but with some misconceptions about which principle applies to the flaw shown.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	**25%**	Clearly explains why the flaw is misleading, including what a reader could wrongly conclude as a result — not just that something is 'wrong.'	Explains the flaw with adequate reasoning, covering most of why it misleads a reader.	Identifies the flaw but explanation is generic or only loosely tied to its misleading effect.	Little or no explanation of why the flaw is misleading; the critique is a bare assertion.
Application	**20%**	Proposes a specific, well-justified redesign that directly resolves the identified flaw, correctly applying appropriate chart-selection and design principles.	Proposes a workable redesign that resolves the flaw, with only minor gaps in justification.	Proposes a redesign, but it only partially resolves the flaw or lacks sufficient justification.	Fails to propose a workable redesign, or the redesign does not address the identified flaw.

Organization	**15%**	The response is clearly structured, addressing both the critique and the redesign in a logical sequence with coherent overall flow.	The response is organized and addresses both parts, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	**10%**	Both the critique and the redesign are completely accurate and well-suited to the specific chart and data described.	Mostly accurate, with only minor omissions or a small error in the critique or redesign.	Partially correct; the critique or redesign contains notable inaccuracies or mismatches to the scenario.	Largely incorrect or inappropriate, with major gaps in both the critique and the redesign.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Misleading Pie Charts

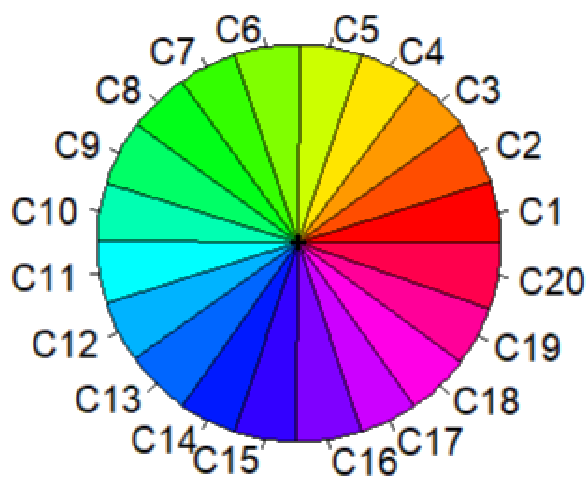
Problem 1 (5 Marks)

Scenario: A customer service report shows a pie chart with 20 nearly-equal-sized slices, one for each complaint category.

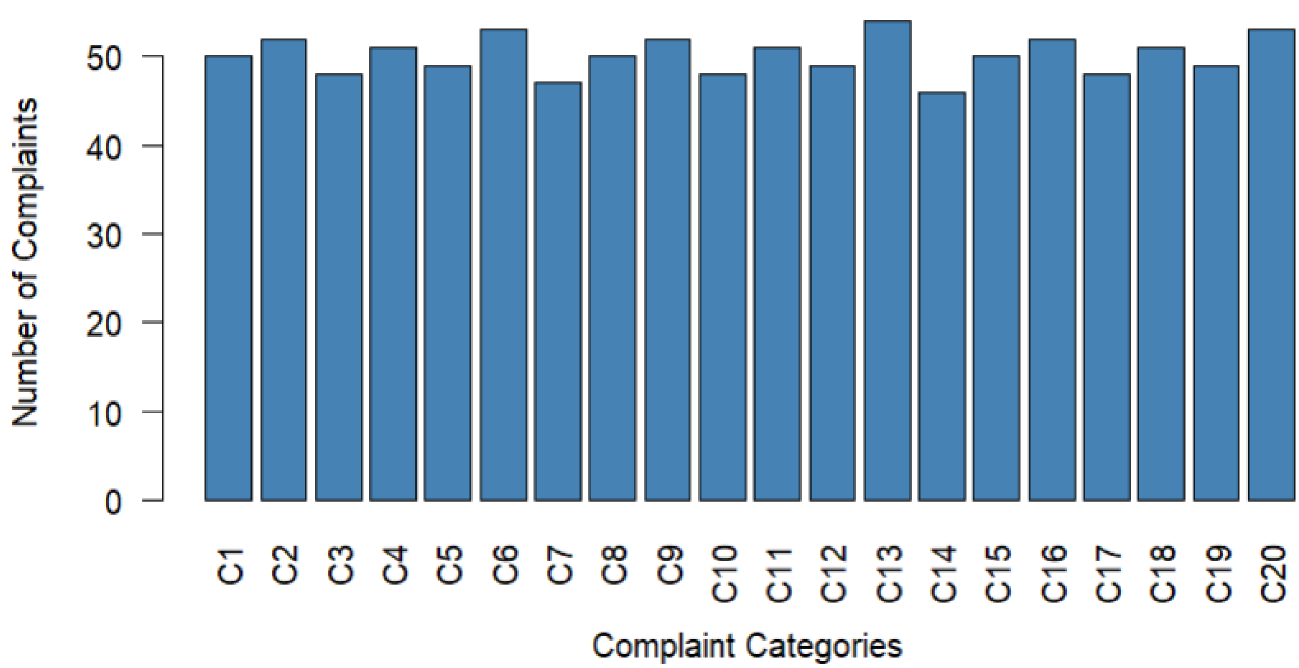
a) Critique this chart: what specific problem does it create for the reader trying to compare categories?

b) Redesign: propose a better chart type for this data, and justify your choice.

Misleading Pie Chart: 20 Complaint Categories



Better Visualization: Bar Chart



a) Critique

A pie chart with 20 nearly equal-sized slices is difficult to read. Small differences between complaint categories are hard to compare because the reader has to compare angles and areas. The large number of slices also makes the chart cluttered.

b) Redesign

A bar chart is better because the lengths of bars can be compared much more accurately. It makes it easy to identify which complaint category has the highest or lowest number of complaints.

RStudio Code

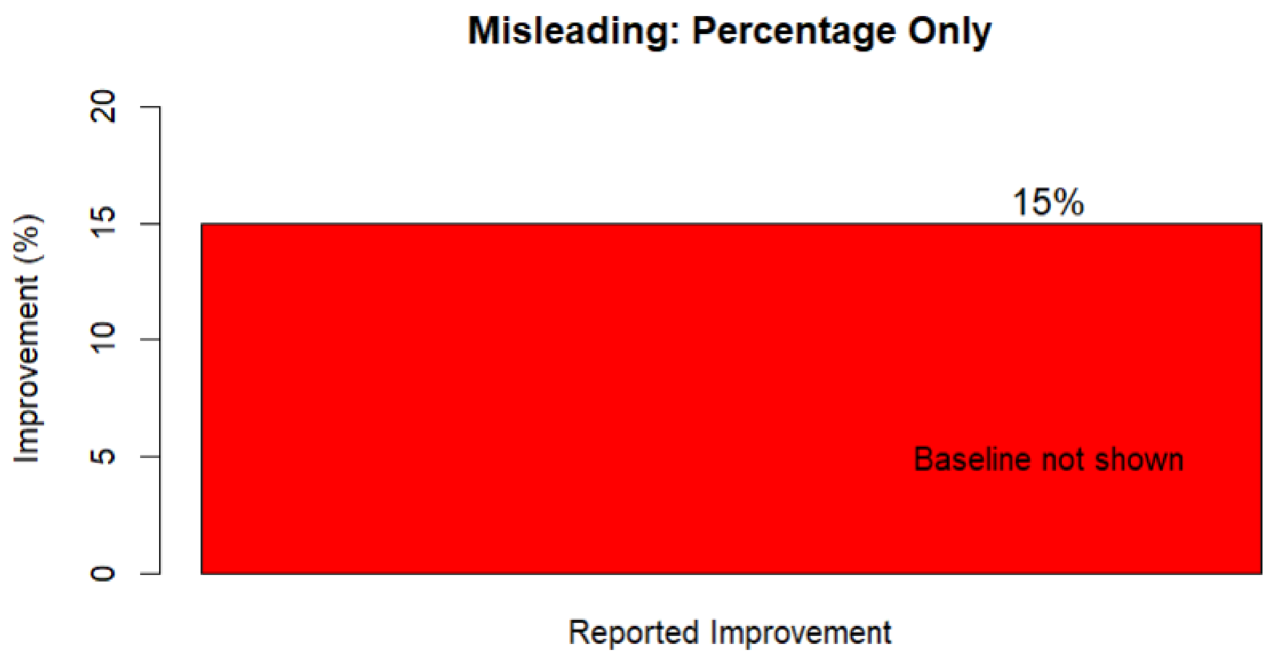
```
categories <- paste0("C", 1:20)

complaints <- c(
  50, 52, 48, 51, 49,
  53, 47, 50, 52, 48,
  51, 49, 54, 46, 50,
  52, 48, 51, 49, 53
)

pie(complaints,
    labels = categories,
    main = "Problem: 20 Nearly Equal Pie Slices")



barplot(complaints,
        names.arg = categories,
        main = "Better: Complaint Categories",
        xlab = "Complaint Category",
        ylab = "Number of Complaints",
        las = 2,
        col = "steelblue")
```

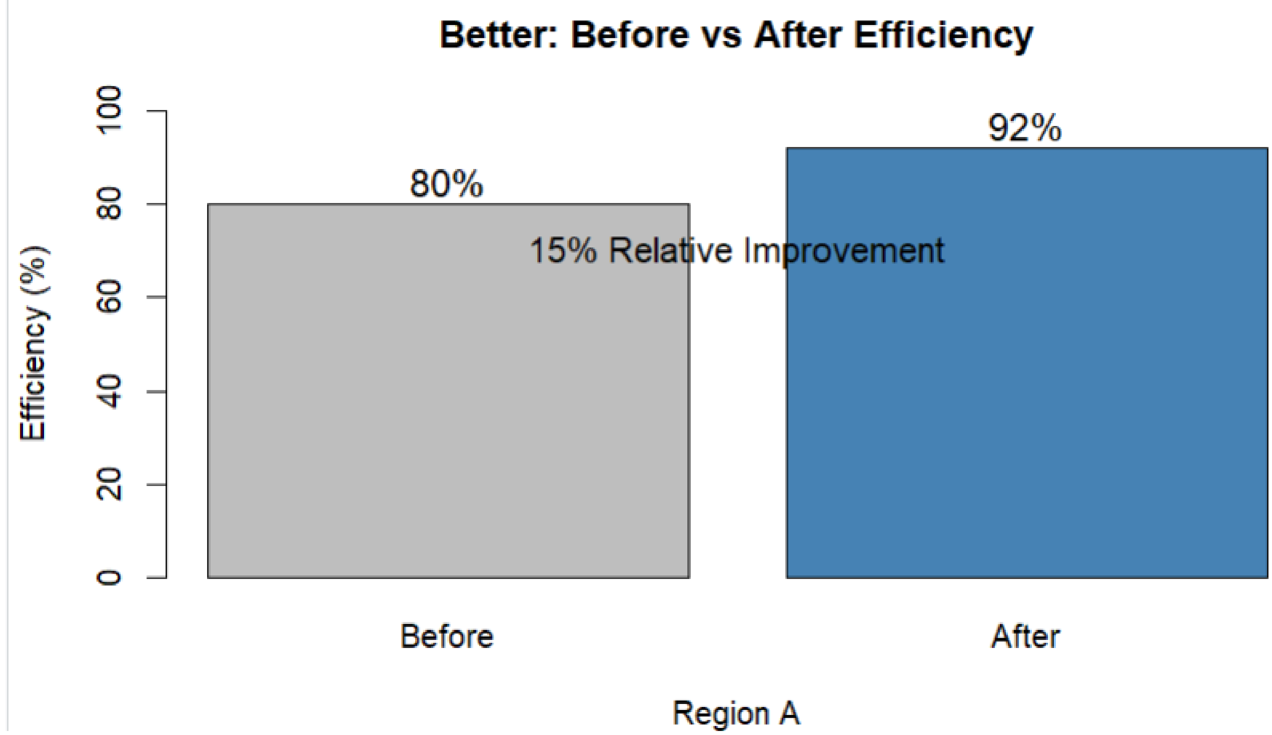
Problem 7 (5 Marks)

Scenario: A city report states "Region A's efficiency improved by 15%" without stating the original baseline value or total scale.

a) Critique why percentage-only reporting like this can be misleading.

b) Redesign: what additional information or chart element would you add to fix this?





a) Critique

The statement **“Region A's efficiency improved by 15%”** is misleading because the **original baseline is missing**.
For example, an increase from 20% to 23% and an increase from 80% to 92% can both represent a 15% relative improvement, but the actual values and practical impact are very different.

b) Redesign

Show the **before value, after value, baseline scale, and percentage change**. A before-and-after bar chart provides the missing context.

RStudio Code – 2 Diagrams

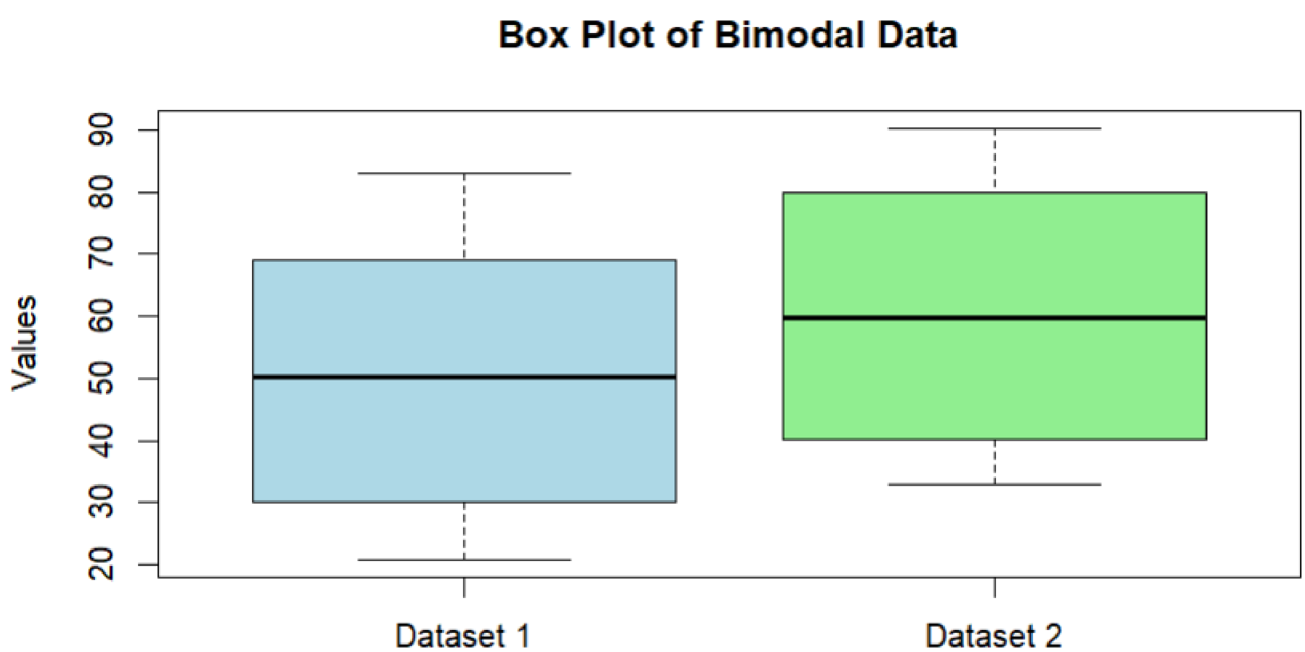
```
before <- 80
after <- 92
barplot(15,
        names.arg = "Improvement",
        ylim = c(0, 20),
        main = "Problem: Percentage Only",
        ylab = "Reported Improvement (%)",
        col = "red")

text(1, 16,
     "15%",
     cex = 1.3)

text(1, 8,
     "Baseline not shown",
     cex = 1)

efficiency <- c(before, after)

barplot(efficiency,
        names.arg = c("Before", "After"),
        ylim = c(0, 100),
        main = "Better: Region A Efficiency",
        ylab = "Efficiency (%)",
        col = c("gray", "steelblue"))

text(c(0.7, 1.9),
     efficiency + 5,
     labels = paste0(efficiency, "%"),
     cex = 1.2)

text(1.3, 70,
     "15% Relative Improvement",
     cex = 1.1)
```

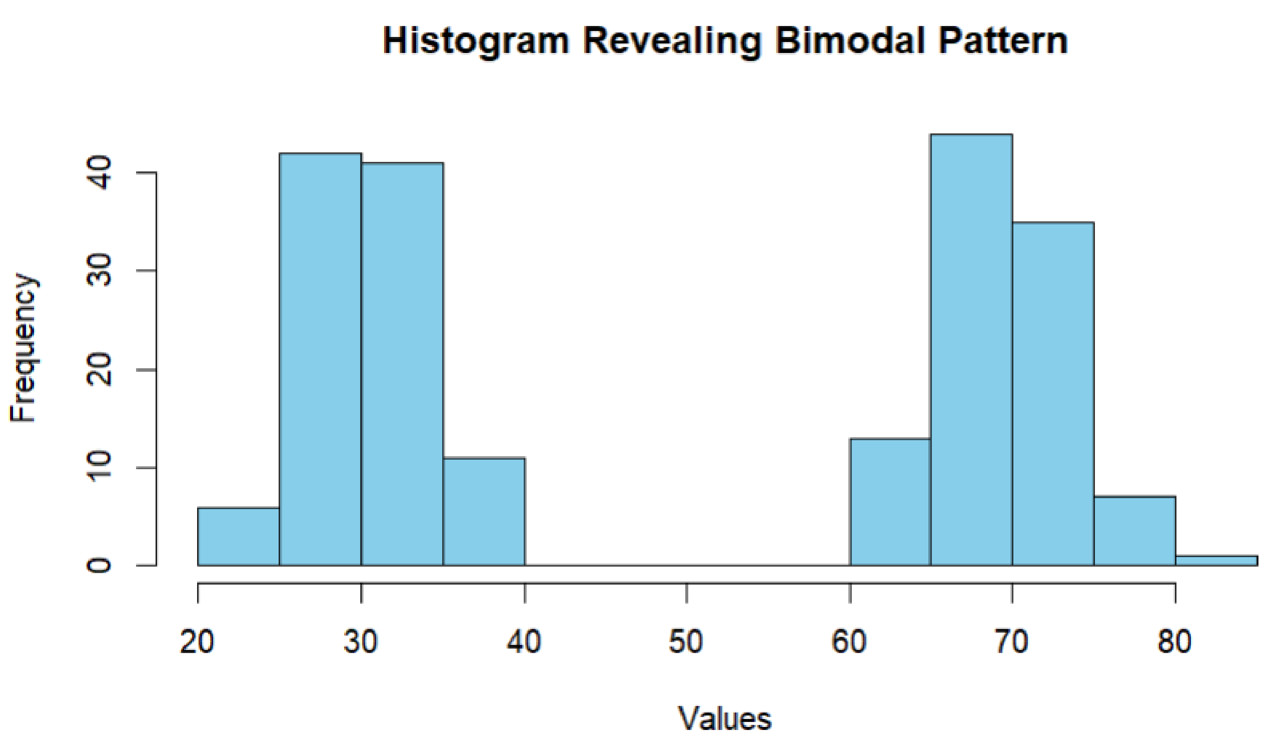
Problem 13 (5 Marks)

Scenario: A box plot is used to compare two datasets that are later revealed to be bimodal (two distinct peaks), but the box plot shows no visible sign of this.

a) Critique what limitation of box plots this scenario reveals.

b) Redesign: propose an alternative chart that would reveal the bimodal pattern.





Problem 13 – Bimodal Data

a) Critique

A box plot summarizes data using the **median, quartiles, and outliers**, but it does not show the detailed shape of the distribution. Therefore, it can hide the fact that the data has **two distinct peaks (bimodal distribution)**.

b) Redesign

Use a **histogram** or density plot. A histogram shows the frequency distribution and makes the **two peaks** visible.

RStudio Code

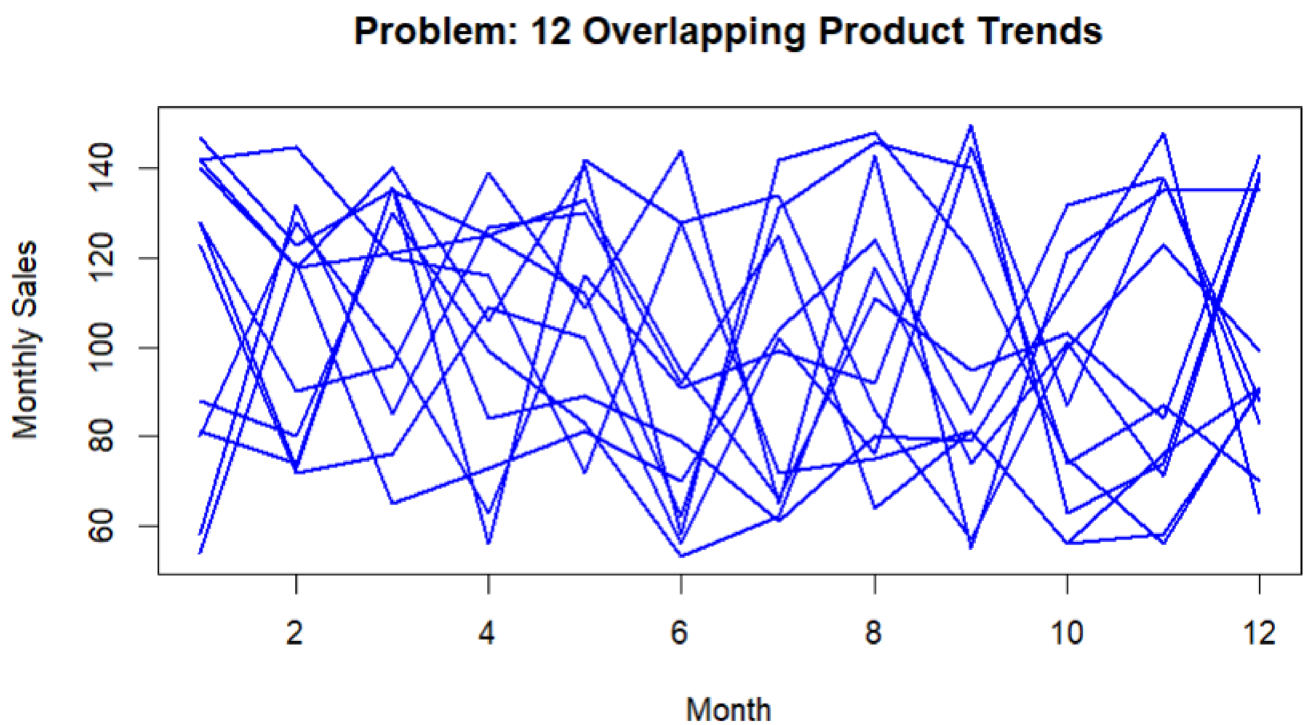
```
set.seed(123)

data1 <- c(
  rnorm(100, mean = 30, sd = 4),
  rnorm(100, mean = 70, sd = 4)
)

boxplot(data1,
        main = "Problem: Box Plot of Bimodal Data",
        ylab = "Values",
        col = "lightblue")


hist(data1,
     breaks = 20,
     main = "Better: Histogram Revealing Bimodal Pattern",
     xlab = "Values",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")
```

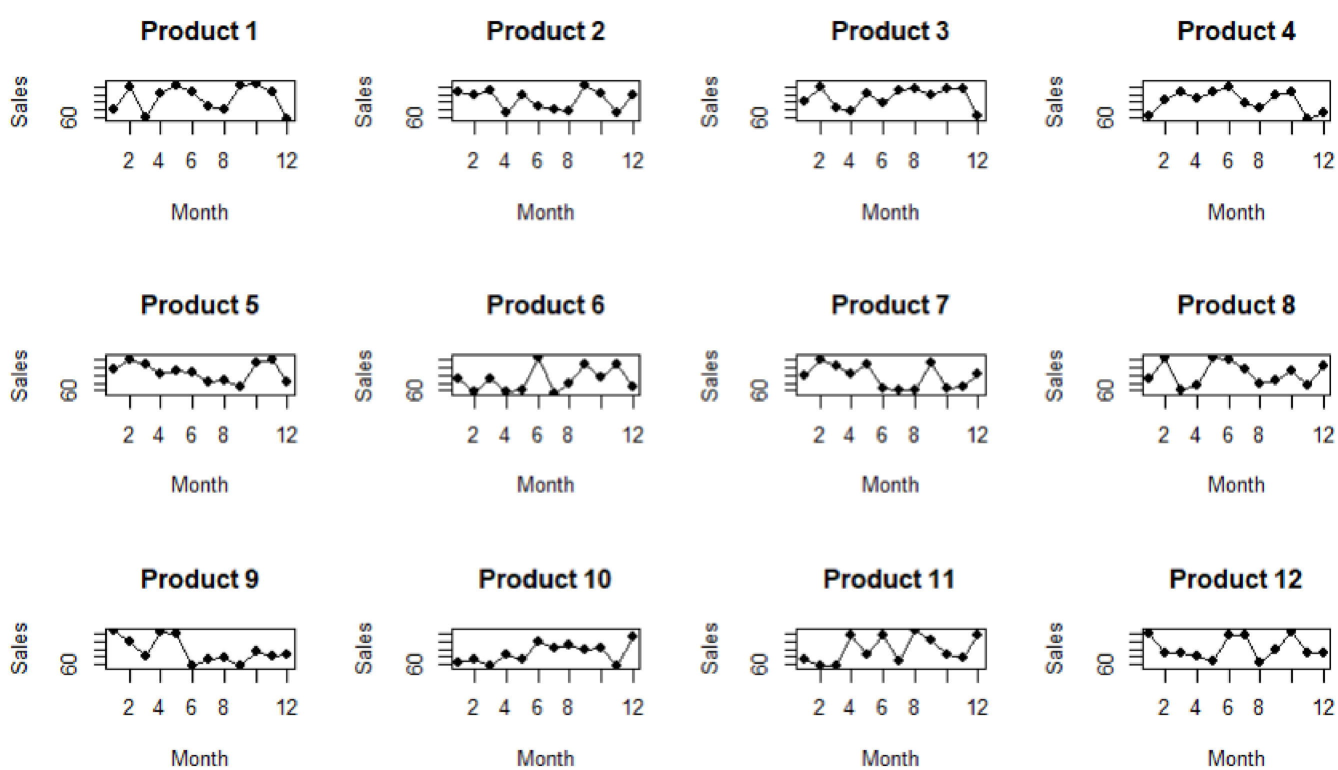
Problem 19 (5 Marks)

Scenario: A report shows 12 overlapping line charts (one per product), all drawn in similar shades of blue with no legend, comparing monthly sales trends.

a) Critique the specific problems with this chart.

b) Redesign: propose a clearer way to present 12 products' trends.





a) Critique

The chart has several problems:

- **12 lines overlap**, making individual trends difficult to follow.
- Similar **blue shades** make the products difficult to distinguish.
- There is **no legend**, so the reader cannot identify each line.
- The chart becomes visually cluttered.
- Important increases and decreases can be hidden by other lines.

b) Redesign

Use **small multiple line charts**, with one separate chart for each product. Keeping the same scale makes comparison possible while avoiding overlapping lines.

RStudio Code

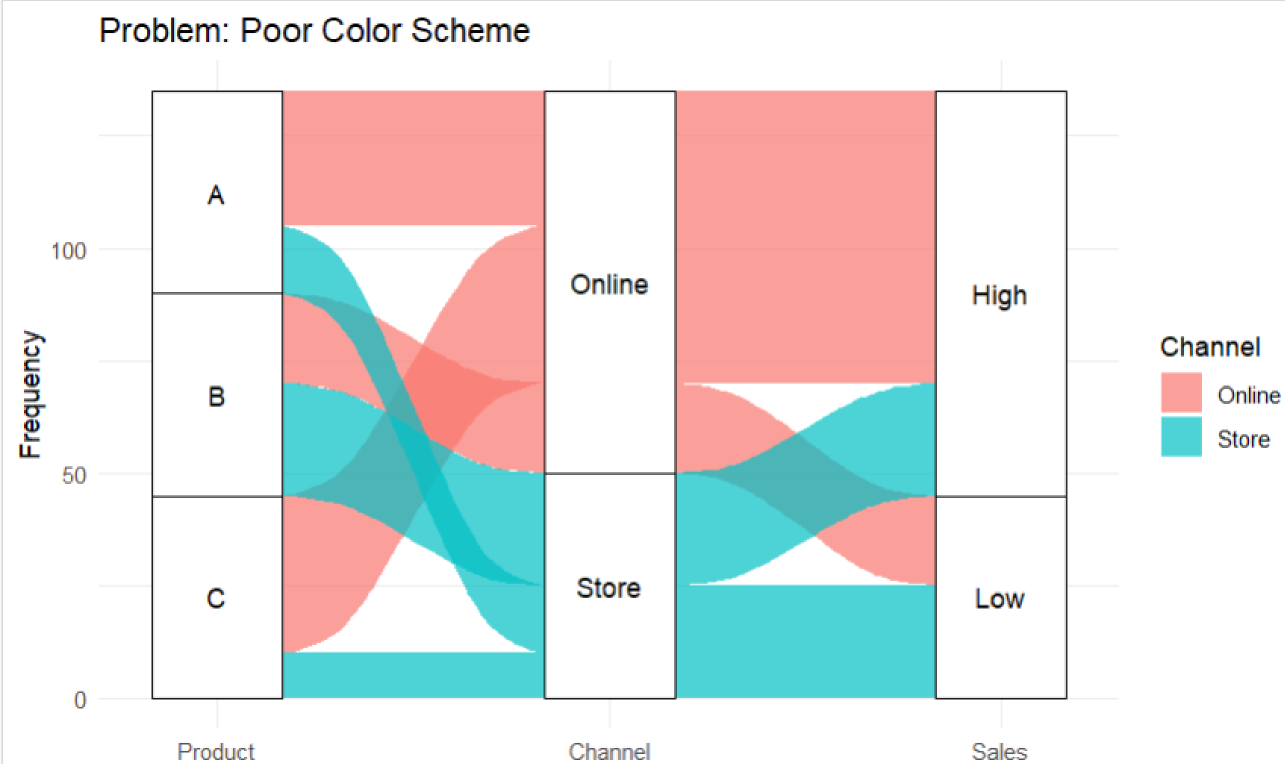
```
set.seed(123)

months <- 1:12

sales <- matrix(
  sample(50:150, 12 * 12, replace = TRUE),
  nrow = 12,
  ncol = 12
)
matplot(months,
        sales,
        type = "l",
        lty = 1,
        lwd = 2,
        col = "blue",
        main = "Problem: 12 Overlapping Product Trends",
        xlab = "Month",
        ylab = "Monthly Sales")

par(mfrow = c(3, 4))

for (i in 1:12) {

  plot(months,
       sales[i, ],
       type = "o",
       pch = 16,
       main = paste("Product", i),
       xlab = "Month",
       ylab = "Sales",
       ylim = range(sales))
}

par(mfrow = c(1, 1))
```

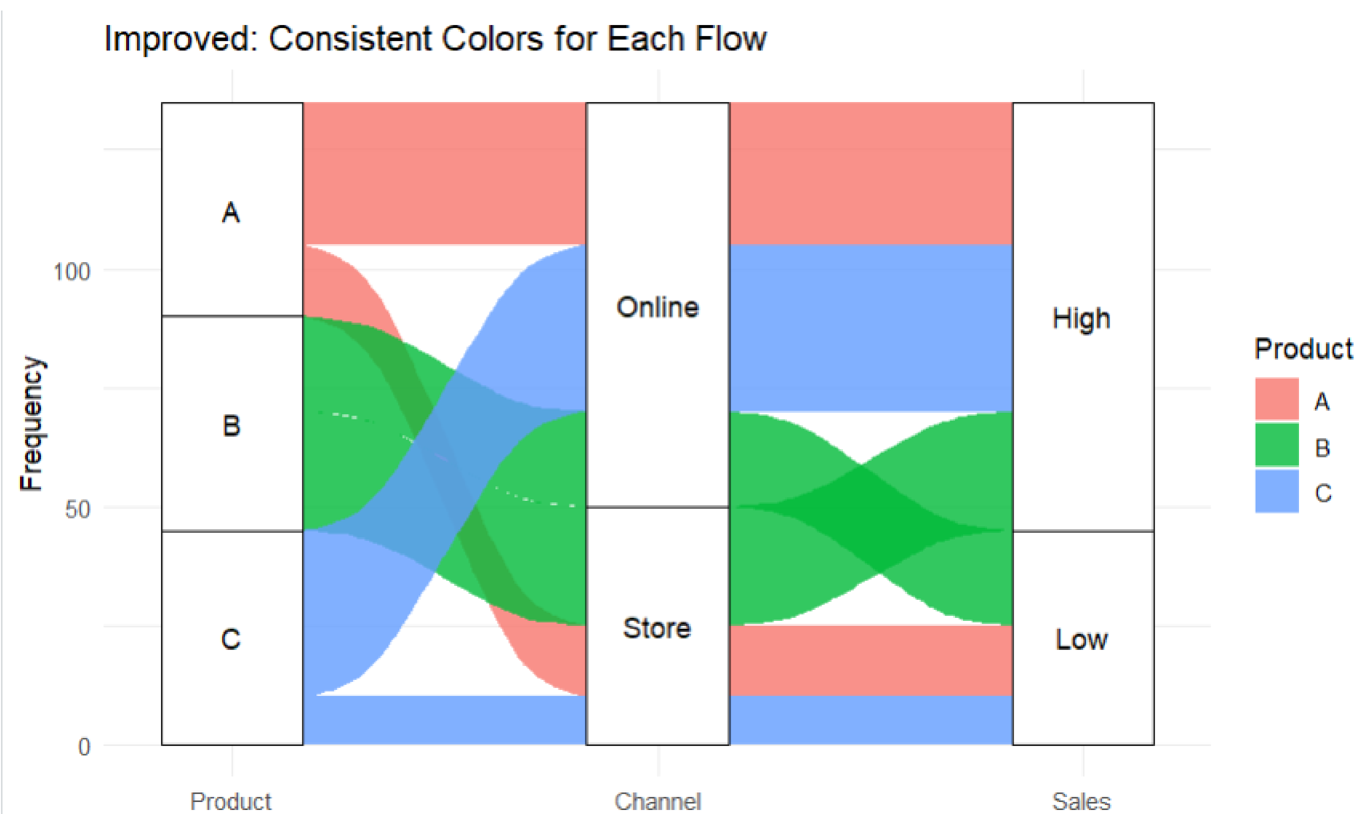
Problem 25 (5 Marks)

Scenario: A parallel sets diagram connecting 3 categorical variables uses the same color for several unrelated categories, making it hard to trace any single ribbon through the diagram.

a) Critique this specific design flaw.

b) Redesign: propose how you would fix the color scheme to make the flows traceable.





a)Critique

Using the **same color for unrelated categories** makes it difficult to trace a ribbon through the three categorical variables. The viewer may incorrectly assume that ribbons with the same color belong to the same category or flow.

b) Redesign

Give each meaningful category or flow a **distinct and consistent color**. The same color should remain attached to that flow across all three variables. A clear legend should also be provided.

RStudio Code

```
library(ggplot2)
library(ggalluvial)

data <- data.frame(
  Product = c("A", "A", "B", "B", "C", "C"),
  Channel = c("Online", "Store",
            "Online", "Store",
            "Online", "Store"),
  Sales = c("High", "Low",
          "Low", "High",
          "High", "Low"),
  Frequency = c(30, 15, 20, 25, 35, 10)
)
ggplot(data,
      aes(axis1 = Product,
          axis2 = Channel,
          axis3 = Sales,
          y = Frequency)) +

  geom_alluvium(aes(fill = Channel),
              alpha = 0.7) +

  geom_stratum() +

  geom_text(stat = "stratum",
          aes(label = after_stat(stratum))) +

  scale_x_discrete(
    limits = c("Product", "Channel", "Sales"),
    expand = c(.1, .1)
  ) +

  labs(
    title = "Problem: Poor Color Scheme",
    y = "Frequency"
  ) +

  theme_minimal()
```

```
ggplot(data,
      aes(axis1 = Product,
          axis2 = Channel,
          axis3 = Sales,
          y = Frequency)) +

  geom_alluvium(aes(fill = Product),
               alpha = 0.8) +

  geom_stratum() +

  geom_text(stat = "stratum",
           aes(label = after_stat(stratum))) +

  scale_x_discrete(
    limits = c("Product", "Channel", "Sales"),
    expand = c(.1, .1)
  ) +

  labs(
    title = "Better: Consistent Colors for Each Product",
    y = "Frequency",
    fill = "Product"
  ) +

  theme_minimal()
```